

Guiding users to auto-processing results

... via metadata from multi-sweep experiments

Clemens Vonnrhein, Global Phasing Ltd.
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Introduction

The Global Phasing Workflow (GPhL WF) is now deployed at various beamlines - in production, near-production or late testing modes.

Most recent work was done on MASSIF-1 in collaboration with Matt Bowler and Olof Svensson (ESRF).

A recent session on MASSIF-1 involved

- 6 crystals
- GPhL WF collections of 4 sweeps for 5 of those and a single sweep for the other
- 21 sweeps in total

The ESRF Data Portal (<https://data.esrf.fr/>) is one of the most advanced and modern interfaces an MX beamline user will see - to access the data collected and processed during a session.

4-Sweep GPhL WF collection

18/05/2026 16:39 started at 16:21 TLN-x8

18/05/2026 16:21:15 MXPress Global Phasing Advanced Strategy

Level 1

Kappa settings : 5

Best auto processing ?

From autoPROC

Merged

Orthorhombic system (C2221)

a=82.0 Å b=111.6 Å c=62.7 Å

	Compl.	Res. low	Res. high	Rmeas	<I/s(I)>	cc1/2	ccAno
inner	100.00%	45.47	3.47	8.80	36.00	1.00	-
outer	97.10%	1.30	1.28	230.60	1.00 ★	0.43	-
overall	99.90%	45.47	1.28	13.30	14.20	1.00	-

4-sweep data collection,
single processing with autoPROC
(via aP_wf_process interface)

Still a few wrinkles (but nearly
there)

Auto processing ranking ?

Program	a,b,c (Å)	α,β,γ (°)	Compl.	Res. low	Res. high	Rmeas	<I/s(I)>	cc1/2	ccAno
18/05/2026 16:40									
autoPROC	82.0	90.0	inner	100.00%	45.47	3.47	8.80	36.00	1.00
C2221	111.6	90.0	outer	97.10%	1.30	1.28	230.60	1.00 ★	0.43
Merged	62.7	90.0	overall	99.90%	45.47	1.28	13.30	14.20	1.00
18/05/2026 16:48									
autoPROC	82.0	90.0	inner	100.00%	45.47	3.47	8.80	36.00	1.00
C2221	111.6	90.0	outer	97.10%	1.30	1.28	230.60	1.00 ★	0.43
Merged	62.7	90.0	overall	99.90%	45.47	1.28	13.30	14.20	1.00

**Level 2 (list of individual
processing results)**

06_icat_astra

Summary Input Files

/mnt/multipath-shares/data/visitor/blc16980/id30a1/20260331/PROCESSED_DATA/TLN/TLN-x8/run_01_MXPressG/TLN-x8_1_001/main/06_icat_astra

Download (669.45 MB)

Search for a file

.../1_001/main/06_icat_astra/online

- summary.html
- summary.tar.gz
- truncate-unique.mtz
- truncate-unique.stats
- truncate-unique.table1

1 MTZ file

Level 3

06_icat_astra_staraniso

Summary Input Files

/mnt/multipath-shares/data/visitor/blc16980/id30a1/20260331/PROCESSED_DATA/TLN/TLN-x8/run_01_MXPressG/TLN-x8_1_001/main/06_icat_astra_staraniso

Download (673.01 MB)

Search for a file

.../06_icat_astra_staraniso/online

- staraniso_alldata-unique.mtz
- staraniso_alldata-unique.stats
- ...staraniso_alldata-unique.table1
- summary.html
- summary.tar.gz

1 MTZ file

Level 3

4-Sweep GPhL WF collection

18/05/2026 16:39 started at 16:21 TLN-x8

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... Best auto processing ?

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	Compl.	Res. low	Res. high	Rmeas	<I/s(I)>	cc1/2	ccAno
inner	100.00%	45.47	3.47	8.80	36.00	1.00	-
outer	97.10%	1.30	1.28	230.60	1.00 ★	0.43	-
overall	99.90%	45.47	1.28	13.30	14.20	1.00	-

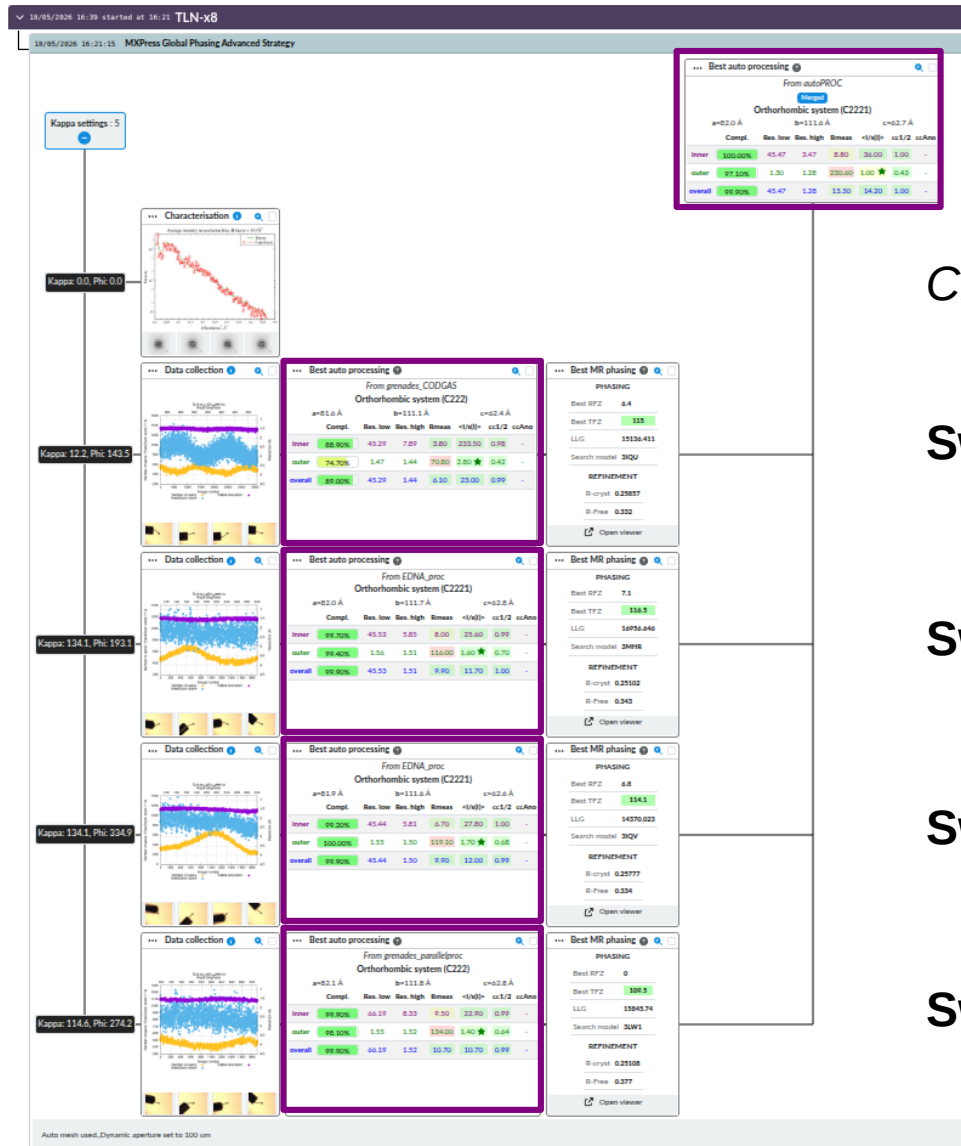
Auto mesh used., Dynamic aperture set to 100 um

expanding the "Kappa settings: <N>" item gives us a view similar to single sweep datasets

4-Sweep GPhL WF collection

Everything shown at Level 1

single dataset from underlying
list of processing results



Characterisation + Strategy

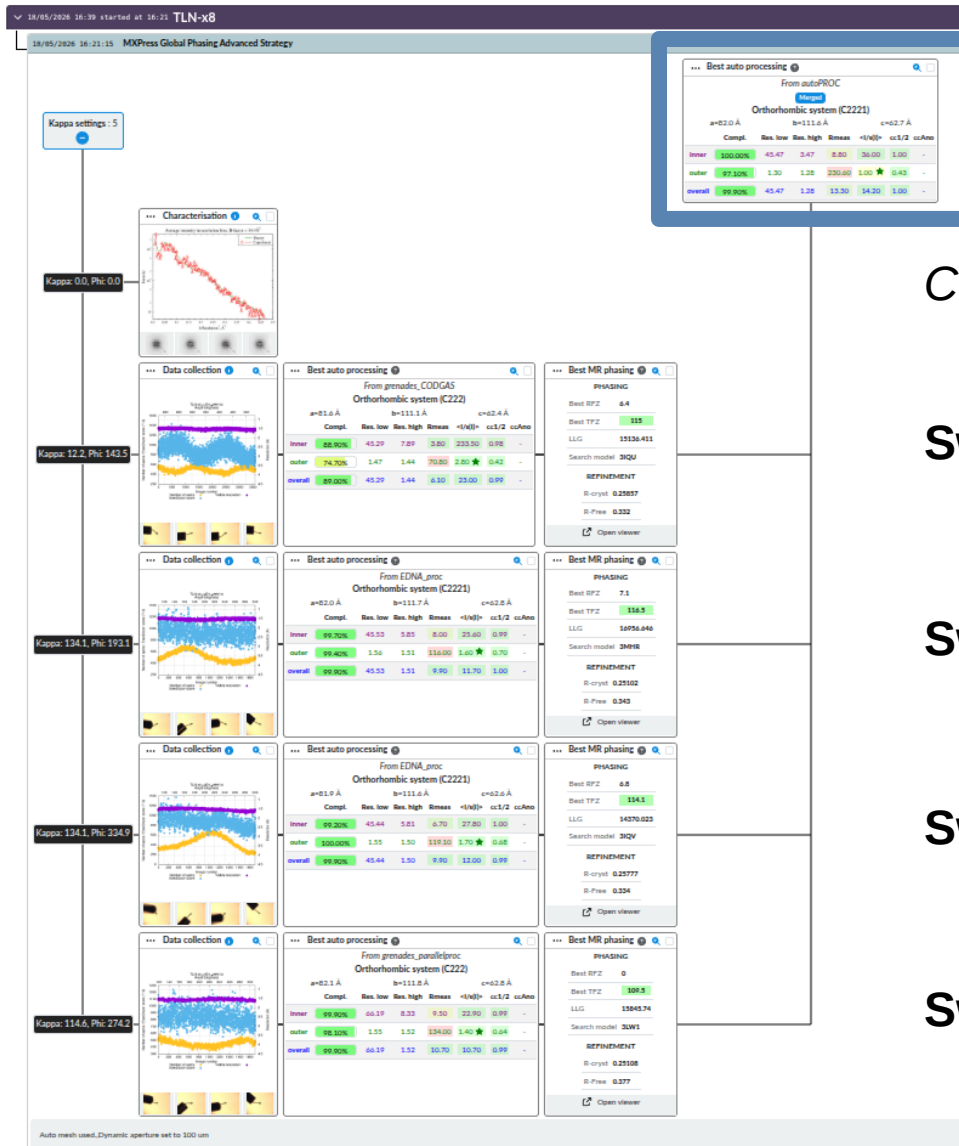
Sweep 1

Sweep 2

Sweep 3

Sweep 4

4-Sweep GPhL WF collection



Combined processing, GPhL WF

Characterisation + Strategy

Sweep 1

automatic data processing
as single sweep
with all available pipelines

Sweep 2

automatic data processing
as single sweep
with all available pipelines

Sweep 3

automatic data processing
as single sweep
with all available pipelines

Sweep 4

automatic data processing
as single sweep
with all available pipelines

User: "What MTZ file should I take?"

Expert (1990): "Have a look at the diffraction spots and if they look nice. What about spatial overlap? Mask your beamstop carefully. Pick some spots for indexing - maybe only the first few images. Check visually that all your spots are predicted. Edit the shoebox and profile parameters to catch enough background. Integrate your 90 degree of data and look at Rmerge, $\langle I/\sigma I \rangle$ and completeness. Calculate some isomorphous Patterson maps: do you have enough anomalous or isomorphous signal to find your heavy atoms on Harker sections? Have you done the self-rotation computation for NCS averaging later?"

Expert (2000): "Look at the interleaved wavelength and/or inverse-beam MAD dataset that gives you the strongest anomalous signal. Resolution is not that important since you first need to solve the structure (and collect a high-res native later anyway). See if the peak SAD dataset gives you a better set of initial HA sites. Run SHELXD over the weekend - you might only get 1 solution in a million."

Expert (2010): "Have a look at those with the highest resolution, a good Rmeas or Rpim value in the outer shell, sensible completeness and $\langle I/\sigma I \rangle$ value. For SAD data look at the anomalous signal. Be careful about radiation damage (this is why we collect high-multiplicity data). Try different pipelines."

Expert (2020): "Take the best one(s) and run MR against all possible AF models or the whole PDB - and take the best result after that."

Expert (2026): "Here it is."

User: "What MTZ file should I take?"

Expert (1990): "Have a look at the diffraction spots and if they look nice. What about spatial overlap? Mask your beamstop carefully. Pick some spots for indexing - maybe only the first few images. Check visually that all your spots are predicted. Edit the shoebox and profile parameters to catch enough background. Integrate your 90 degree of data and look at Rmerge, $\langle I/\sigma I \rangle$ and completeness. Calculate some isomorphous Patterson maps: do you have enough anomalous or isomorphous signal to find your heavy atoms on Harker sections? Have you done the self-rotation computation for NCS averaging later?"

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Expert (2020): "Take the best one(s) and run MR against all possible AF models or the whole PDB - and take the best result after that."

That's where we are now (or should be) ... more or less ...

Expert (2026): "Here it is."



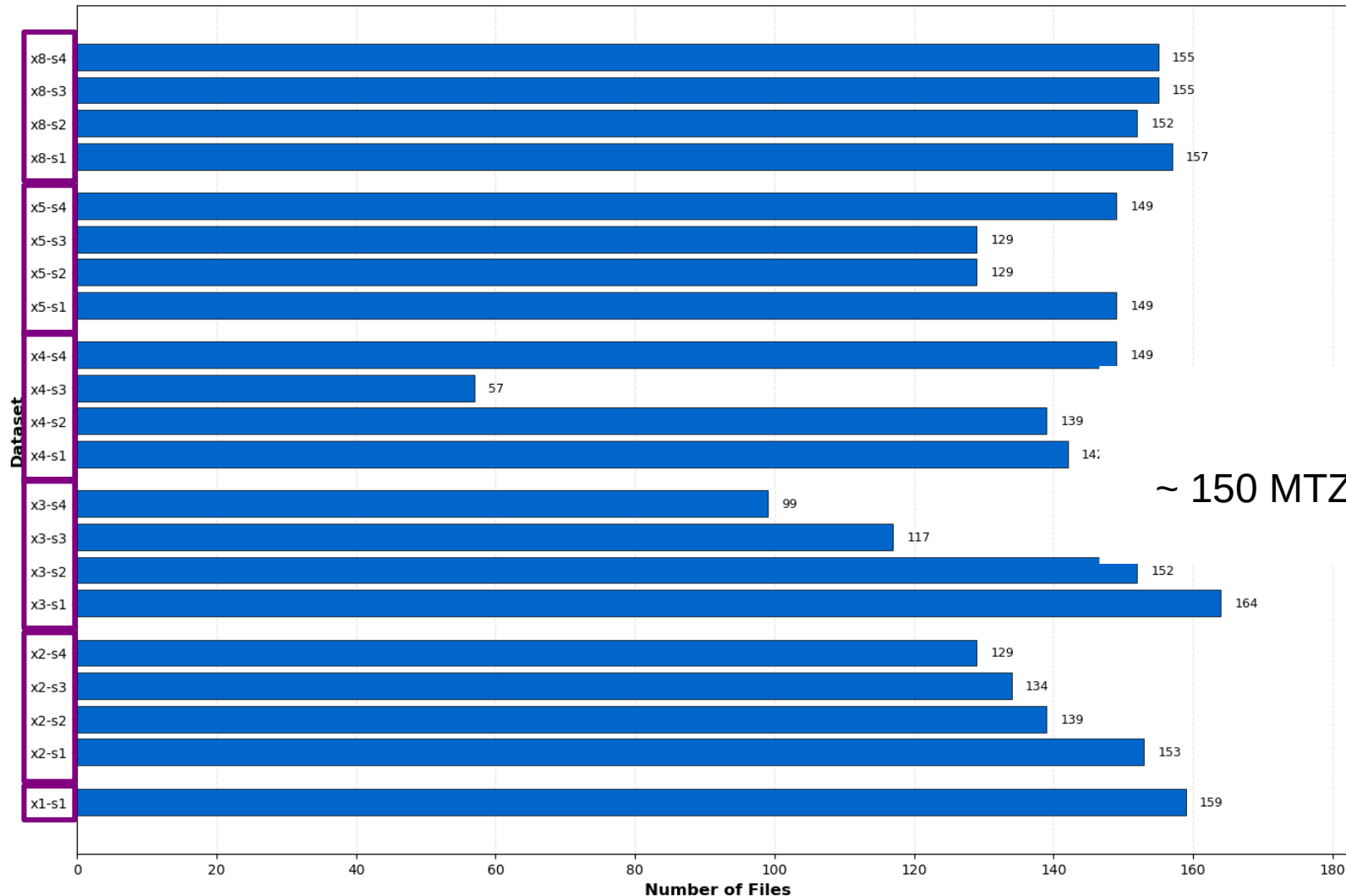
What data have we where?

- All auto-triggered data processing pipelines result in directories/files under the **PROCESSED_DATA** directory
 - typically triggered on a per-sweep basis
- We can easily **rsync** the whole tree
 - before data moves to archive/tape
- Or download **one file at a time** via the web interface
 - from one data processing of one sweep
- Or fetch all files visible in web interface programmatically
 - e.g. via **ICAT+ API**

What MTZ files do we have on disk?

Merged MTZ Files per Dataset (All Files)
(total **2908**)

rsync of PROCESSED_DATA

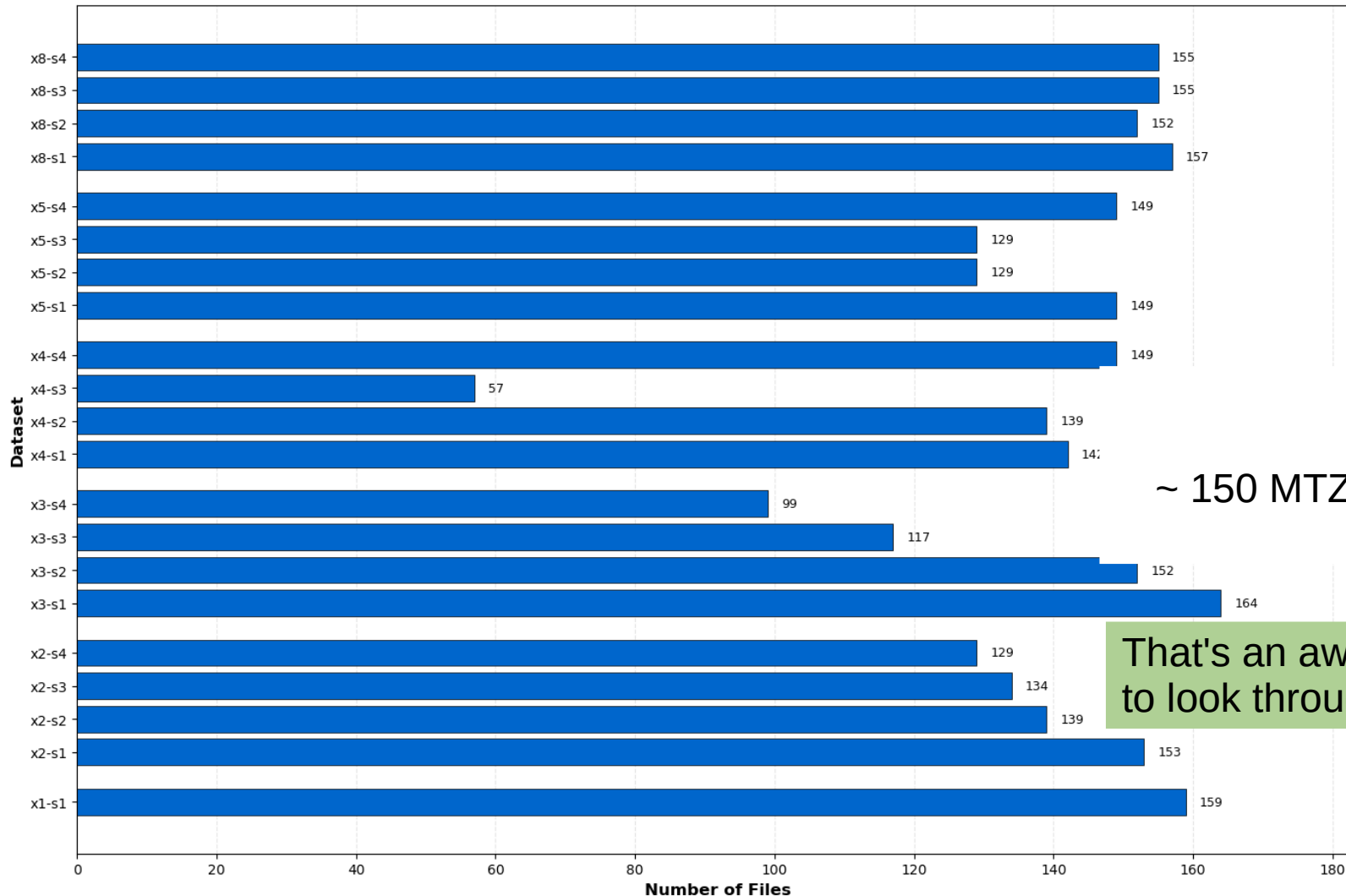


6 crystals
21 sweeps
~ 150 MTZ files / sweep

What MTZ files do we have on disk?

Merged MTZ Files per Dataset (All Files)
(total **2908**)

rsync of PROCESSED_DATA

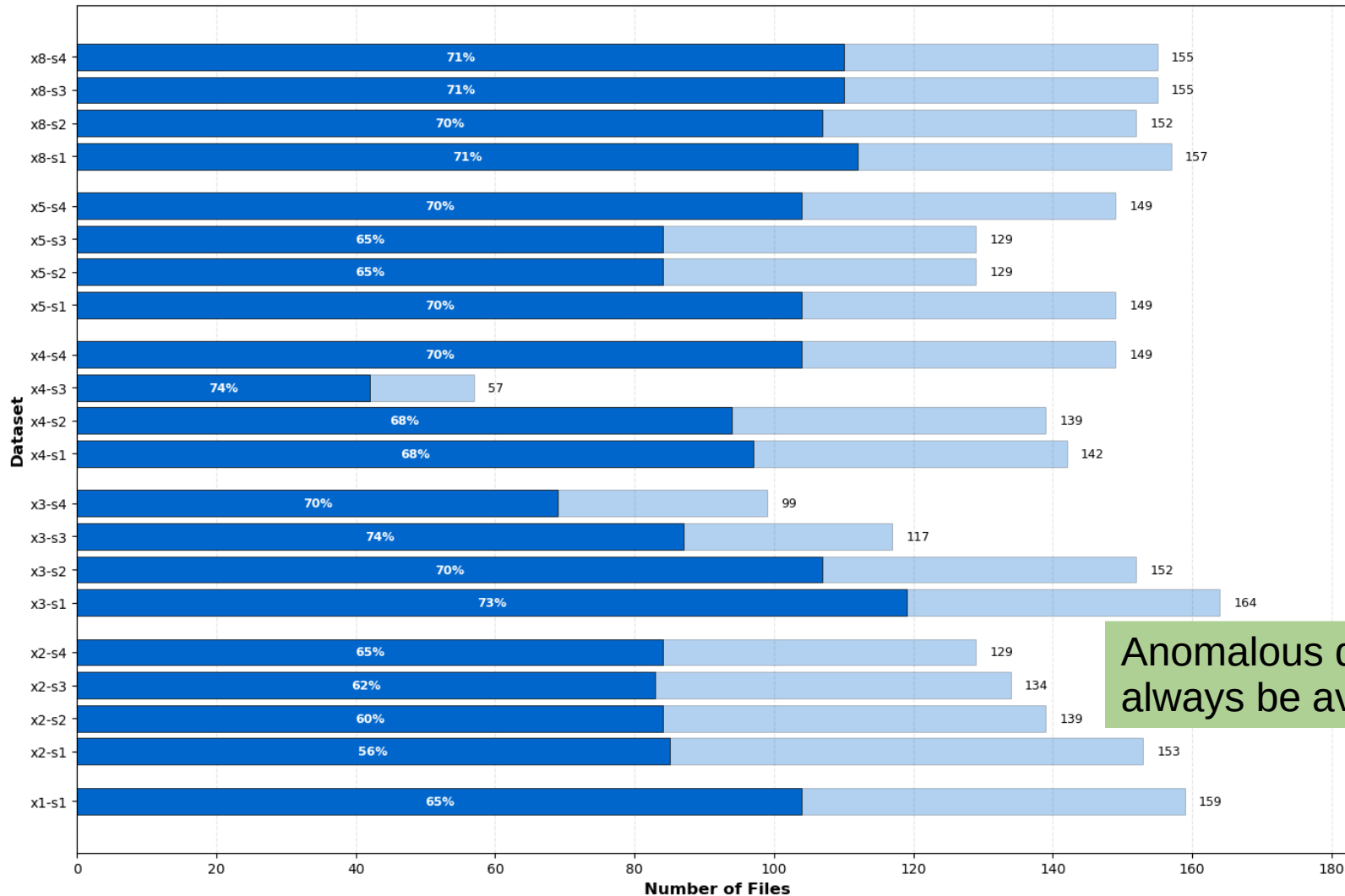


6 crystals
21 sweeps
~ 150 MTZ files / sweep

That's an awful lot of files
to look through ... filter?

What MTZ files do we have on disk?

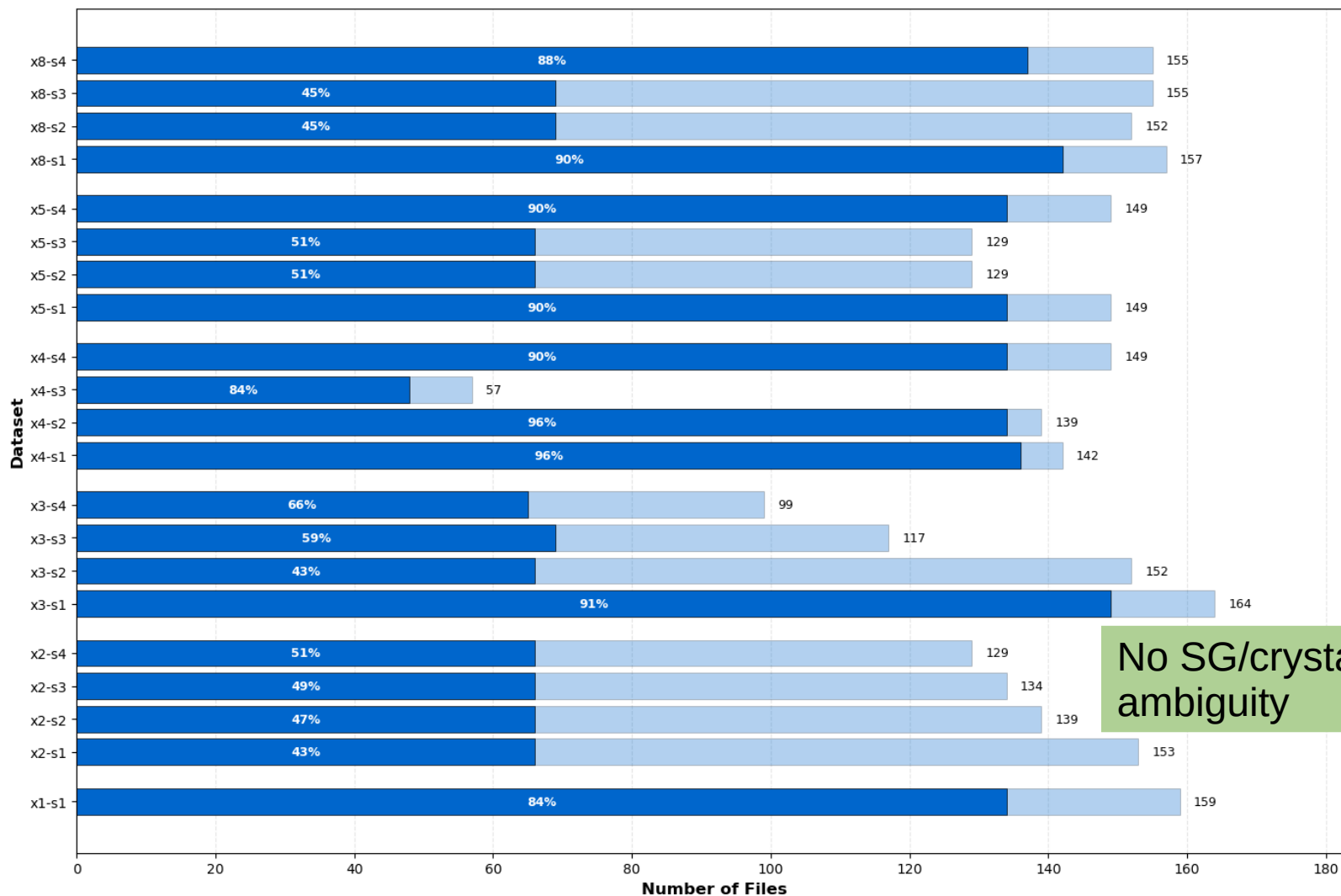
Merged MTZ Files per Dataset (All Files)
only files with anomalous data **1974/2908**



Anomalous data should always be available.

What MTZ files do we have on disk?

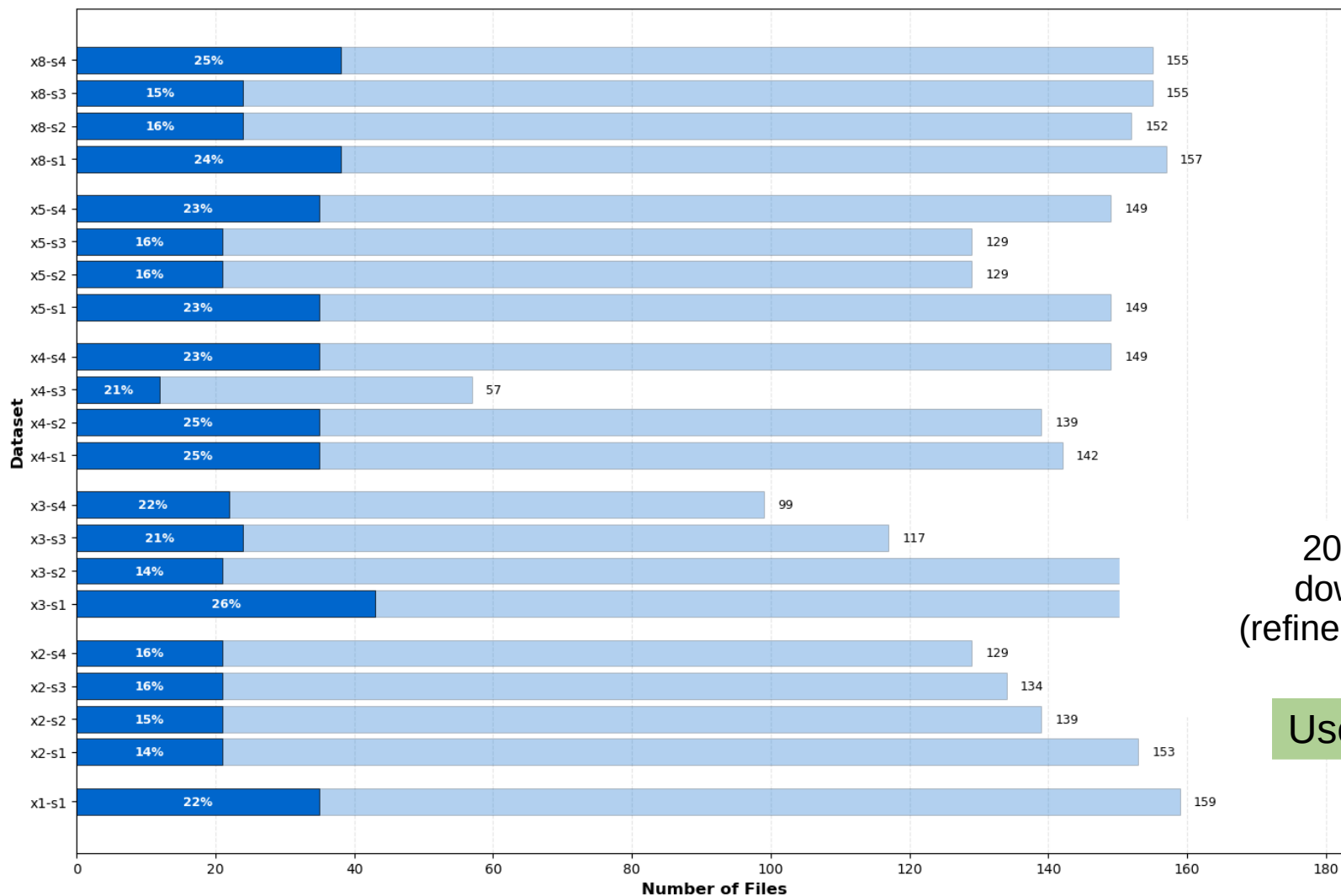
Merged MTZ Files per Dataset (All Files)
only files with SG 20/21 **2016/2908**



No SG/crystal-form
ambiguity

What MTZ files do we have on disk?

Merged MTZ Files per Dataset (All Files)
only with anomalous, excluding phases, SG 20/21, with testset & amplitudes (582/2908)

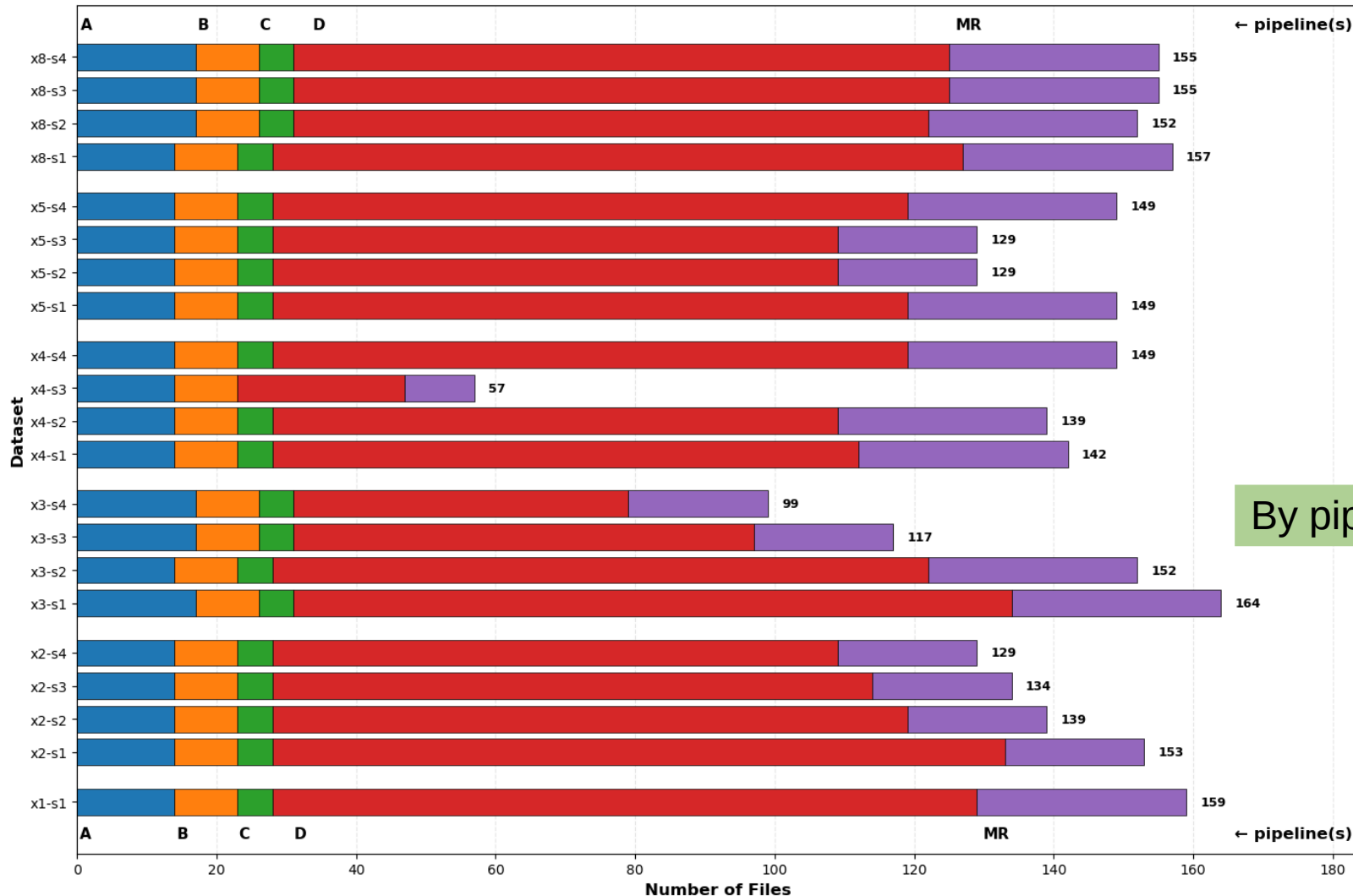


20% data ready for
downstream stages
(refinement, ligand etc)

Useful data

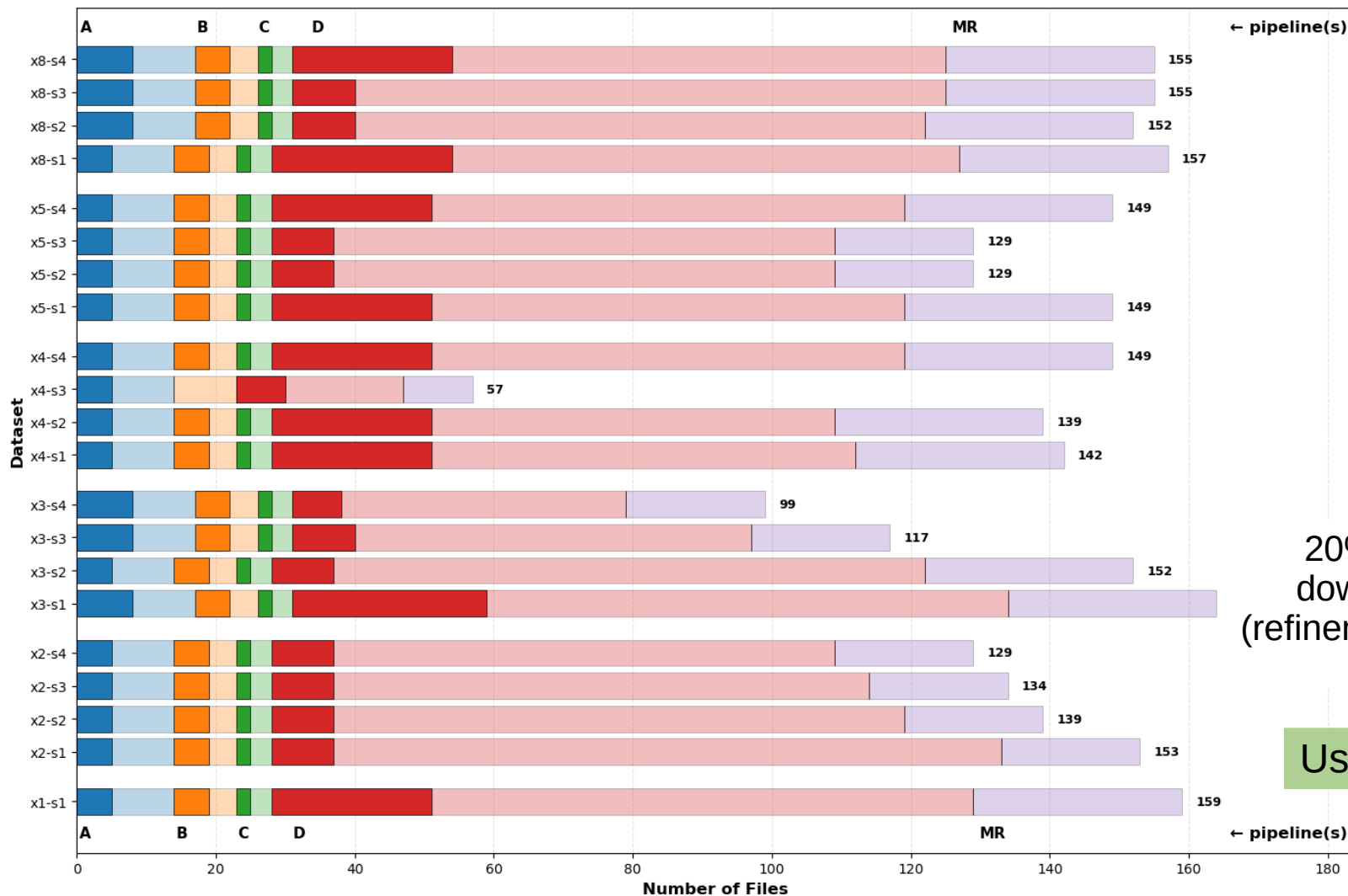
What MTZ files do we have on disk?

Merged MTZ Files per Dataset by Pipeline (All Files)
(total 2908)



What MTZ files do we have on disk?

Merged MTZ Files per Dataset by Pipeline (All Files)
only with anomalous, excluding phases, SG 20/21, with testset & amplitudes (582/2908)



20% data ready for
downstream stages
(refinement, ligand etc)

Useful data

What MTZ files do we have on disk?

- Are users actually rsync'ing the full PROCESSED_DATA over?
- Or are they selecting individual MTZ files manually from the Data Portal?

Auto-processing results, 1 xtal, 4 sweeps

The image displays four screenshots of the 'Auto processing ranking' window, showing search results for different programs. Each window lists programs, their status (inner/outer), and various metrics including Cycles, Res. low, Res. high, Bases, and r/σ. The results are sorted by a ranking metric, with the best results at the top.

Program	Auto (A)	Auto (T)	Cycles	Res. low	Res. high	Bases	r/σ	r/σ ²	r/σ ³
autoPROC_starting	81.8	90.0	inner	30.000	55.73	3.55	1.85	22.60	0.99
XIA2_DIALS	111.5	90.0	outer	62.07	1.28	1.4	1.3	1.3	0.95
gnom3	62.6	90.0	outer	91.60	55.73	1.3	1.3	1.3	0.95

- 58 pipeline runs ...
- ... for one crystal out of 6
- 5 mouse movements and clicks for each
- 290 clicks for a **single crystal**
- Plus giving download file name
 - since pipelines might or might not have created a unique file name
- Will provenance be maintained after such a manual download?
 - Unlikely ...

Auto-processing results, 1 xtal, 4 sweeps

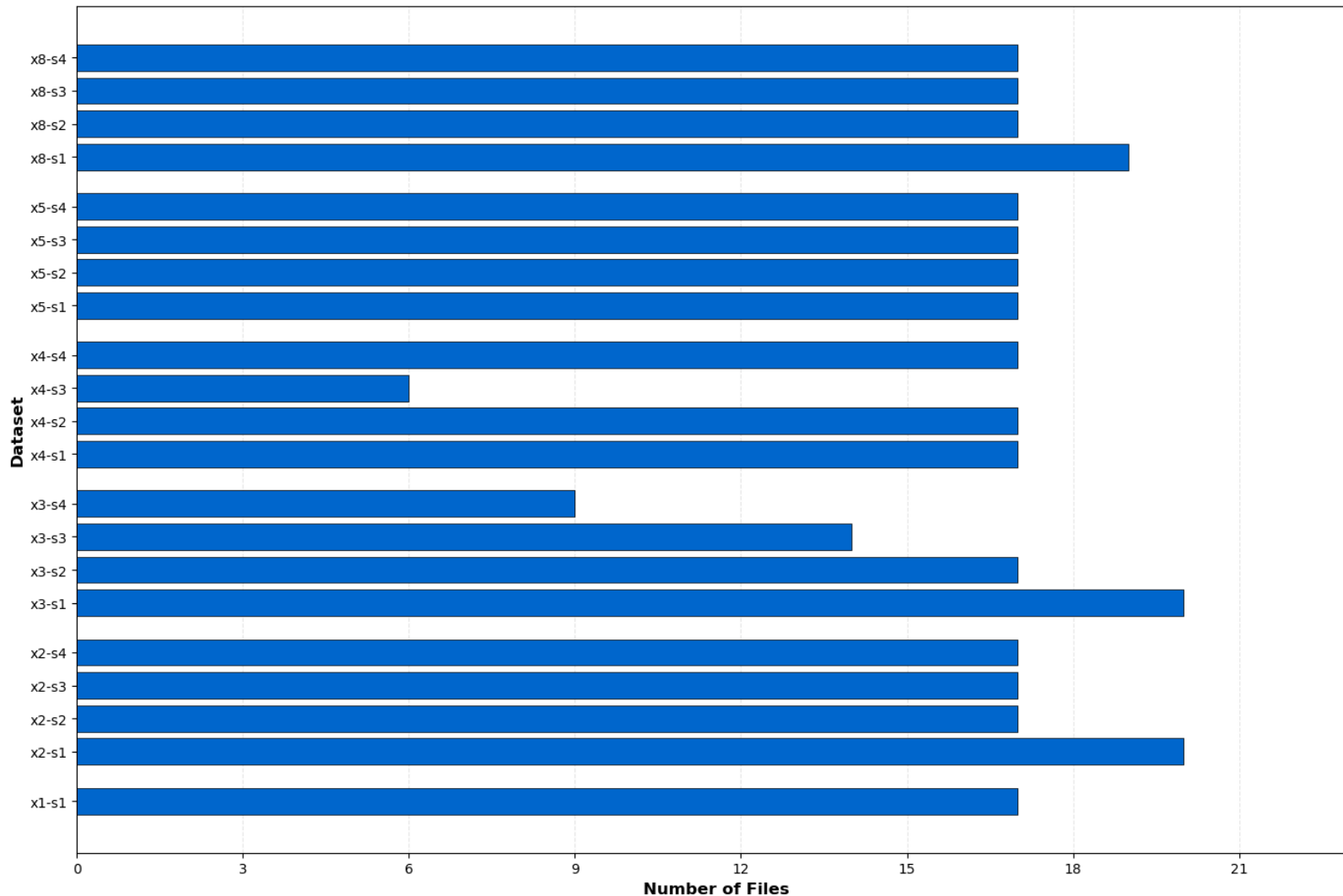
Auto processing ranking											
Program	abs (Å)	rel (Å)	Comp.	Res. low	Res. high	Brewn	r/σ	r/σ ²	cc _{core}		
100000000-10-15	81.8	90.0	inner	300000	55.73	3.55	1.85	22.60	0.99	-	-
autoPROC_starano	111.5	90.0	outer	62070	1.28	1.4	1.3	1.3	202.30	1.40	0.35
C221	62.6	90.0	overall	91.60	55.73	1.3	1.3	1.3	13.30	10.30	1.00
100000000-10-15	81.7	90.0	inner	300000	65.96	3.28	1.57	26.67	0.99	-	-
MAZ_DIALS	111.3	90.0	outer	93070	1.23	1.21	1.05	15.20	0.31	-	-
C221	62.5	90.0	overall	92.07	65.88	1.21	1.21	11.52	9.29	1.00	-
100000000-10-15	81.8	90.0	inner	300000	55.73	3.49	1.95	12.60	0.99	-	-
autoPROC	111.5	90.0	outer	92020	1.21	1.25	1.05	16.30	1.40	0.42	-
C221	62.6	90.0	overall	91.10	55.73	1.25	1.25	13.30	7.20	1.00	-
100000000-10-15	81.8	90.0	inner	99.620	65.95	7.87	10.40	30.00	0.98	-	-
phenix.parallel	111.5	90.0	outer	99.620	1.44	1.44	1.00	1.40	0.56	-	-
C22	62.6	90.0	overall	99.620	65.95	1.44	1.44	11.30	13.30	1.00	-
100000000-10-15	81.8	90.0	inner	99.620	45.37	5.58	1.40	21.20	0.99	-	-
EDNA_gmc	111.4	90.0	outer	300000	1.49	1.44	1.00	1.40	0.42	-	-
C221	62.6	90.0	overall	99.620	45.37	1.44	1.44	11.40	13.30	1.00	-
100000000-10-16	81.8	90.0	inner	99.620	65.91	7.86	10.70	30.00	0.98	-	-
intimed_gmres.fatcrack	111.4	90.0	outer	75.50	1.44	1.44	1.00	1.40	0.42	-	-
C22	62.5	90.0	overall	99.620	65.91	1.44	1.44	11.20	13.40	1.00	-
100000000-10-15	81.8	90.0	inner	99.620	65.91	7.91	11.20	30.00	0.98	-	-
phenix.parallel	111.4	90.0	outer	99.620	1.47	1.44	1.00	1.20	0.51	-	-
C22	62.6	90.0	overall	99.620	65.91	1.44	1.44	11.30	13.20	0.99	-
100000000-10-16	81.6	90.0	inner	100000	45.29	7.89	1.80	23.50	0.98	-	-
phenix.CODAS	111.5	90.0	outer	74.70	1.47	1.44	1.00	1.40	0.42	-	-
C22	62.4	90.0	overall	100000	45.29	1.44	1.44	1.10	23.00	0.99	-
100000000-10-16	81.6	90.0	inner	100000	45.29	7.89	1.80	23.50	0.98	-	-
phenix.CODAS	111.5	90.0	outer	74.70	1.47	1.44	1.00	1.40	0.42	-	-
C22	62.4	90.0	overall	100000	45.29	1.44	1.44	1.10	23.00	0.99	-
100000000-10-15	81.8	90.0	inner	99.700	65.95	7.92	11.20	29.80	0.99	-	-
phenix.parallel	111.5	90.0	outer	99.700	1.47	1.45	1.00	1.40	0.42	-	-
C221	62.6	90.0	overall	99.620	65.95	1.45	1.45	11.40	13.30	1.00	-
100000000-10-15	81.1	90.0	inner	99.500	65.97	8.08	1.40	23.90	0.99	-	-
phenix.parallel	111.5	90.0	outer	99.500	1.50	1.47	1.00	1.20	0.57	-	-
P1	61.1	90.0	overall	99.500	65.97	1.47	1.47	11.00	10.20	0.99	-
100000000-10-15	111.5	90.0	inner	99.500	65.96	8.04	1.70	24.70	0.97	-	-
phenix.parallel	111.5	90.0	outer	99.500	1.49	1.47	1.00	1.40	0.42	-	-
C2	62.6	90.0	overall	99.500	65.96	1.47	1.47	10.80	10.40	0.99	-
100000000-10-15	81.8	90.0	inner	99.100	65.94	8.12	10.10	24.30	0.99	-	-
C2	62.6	90.0	overall	99.100	65.94	1.62	1.62	11.00	10.00	0.99	-
100000000-10-15	82.6	72.6	inner	99.100	65.93	8.29	1.20	19.40	0.97	-	-
phenix.parallel	111.5	90.0	outer	99.100	1.54	1.51	1.00	1.20	0.56	-	-
P1	61.1	89.9	overall	99.100	65.93	1.51	1.51	10.30	8.40	0.99	-
100000000-10-15	82.6	72.6	inner	99.100	65.93	8.29	1.20	19.40	0.97	-	-
phenix.parallel	111.5	90.0	outer	99.100	1.54	1.51	1.00	1.20	0.56	-	-
P1	61.1	89.9	overall	99.100	65.93	1.51	1.51	10.30	8.40	0.99	-
100000000-10-17	82.6	107.4	inner	99.100	65.93	8.29	1.20	19.40	0.97	-	-
phenix.parallel	111.5	90.0	outer	99.100	1.54	1.51	1.00	1.20	0.56	-	-
P1	61.1	89.9	overall	99.100	65.93	1.51	1.51	10.30	8.40	0.99	-

Auto processing ranking											
Program	abs (Å)	rel (Å)	Comp.	Res. low	Res. high	Brewn	r/σ	r/σ ²	cc _{core}		
100000000-10-17	81.8	90.0	inner	300000	66.03	3.41	1.80	25.20	0.99	-	-
MAZ_DIALS	111.4	90.0	outer	74.70	1.28	1.26	1.00	1.40	0.42	-	-
C221	62.6	90.0	overall	91.60	65.94	1.26	1.26	10.60	7.97	0.99	-
100000000-10-17	82.1	90.0	inner	300000	66.14	3.82	1.80	18.90	0.98	-	-
autoPROC_starano	111.7	90.0	outer	93070	1.29	1.4	1.3	1.3	202.30	1.40	0.35
C221	62.8	90.0	overall	91.60	66.14	1.4	1.4	11.30	8.80	0.99	-
100000000-10-17	82.1	90.0	inner	300000	55.86	3.44	1.90	13.40	0.99	-	-
autoPROC	111.7	90.0	outer	92020	1.26	1.34	1.05	16.30	1.50	0.35	-
C221	62.8	90.0	overall	91.60	55.86	1.34	1.34	11.50	6.80	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C22	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
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100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
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C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00	1.30	0.45	-	-
C221	62.8	90.0	overall	99.620	66.14	1.48	1.48	10.20	11.30	0.99	-
100000000-10-17	82.1	90.0	inner	99.620	66.14	6.11	1.40	24.50	0.99	-	-
phenix.parallel	111.7	90.0	outer	99.620	1.51	1.48	1.00				

MTZ files in data portal web interface

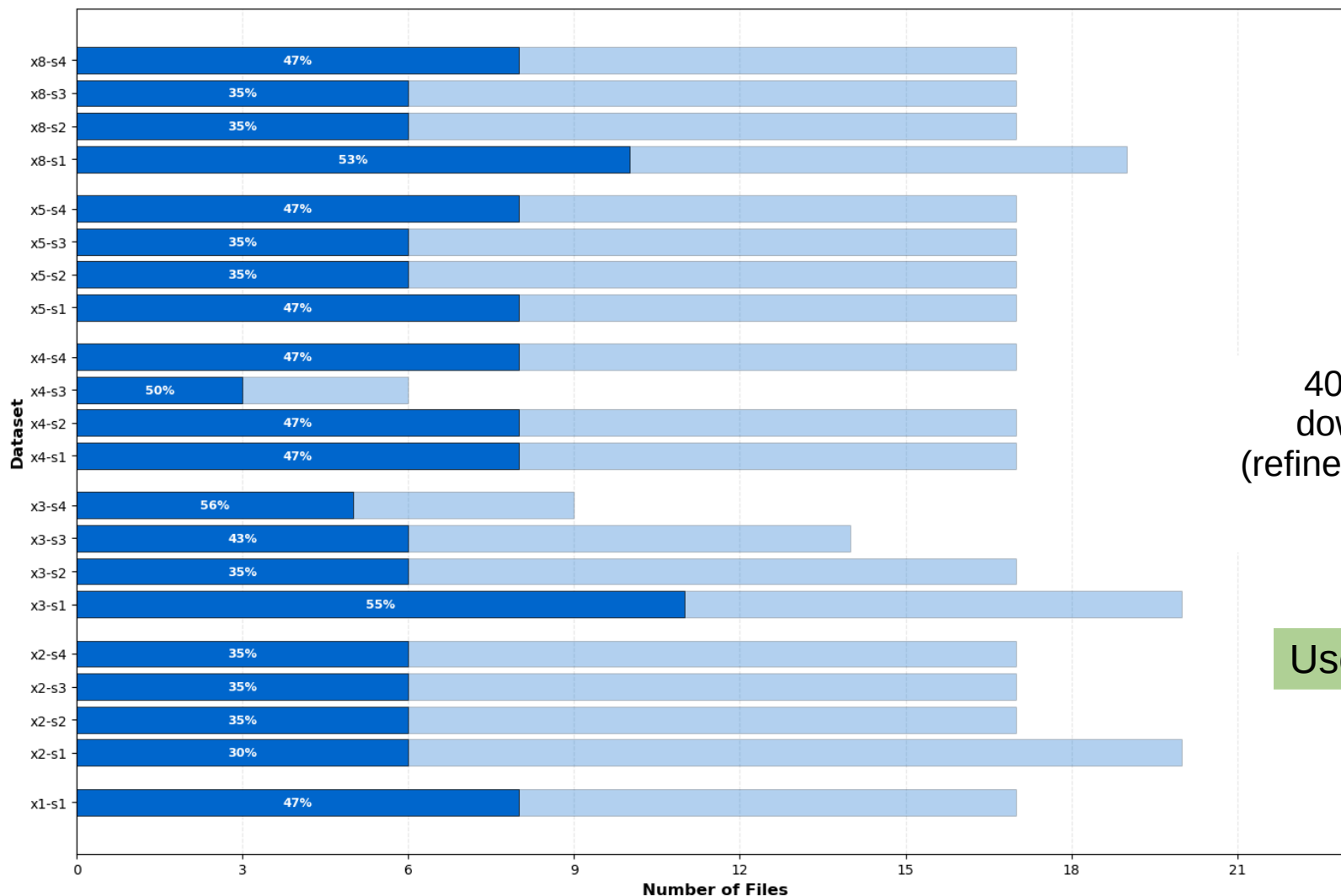
Merged MTZ Files per Dataset (Data Portal)
(total: 343)

via ICAT+ API



MTZ files in data portal web interface

Merged MTZ Files per Dataset (Data Portal)
only with anomalous, excluding phases, SG 20/21, with testset & amplitudes (145/343)

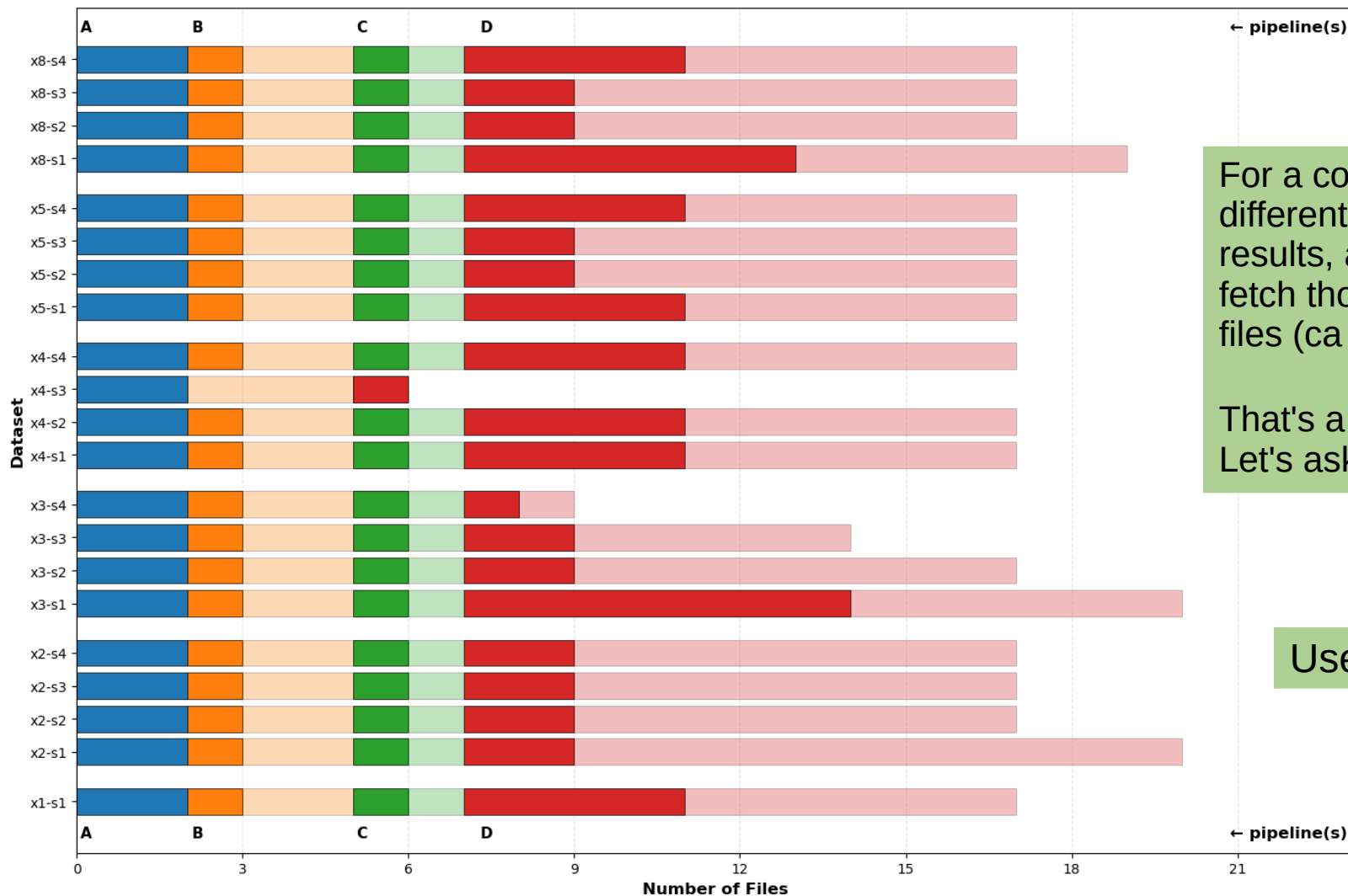


40% data ready for
downstream stages
(refinement, ligand etc)

Useful data

MTZ files in data portal web interface

Merged MTZ Files per Dataset by Pipeline (Data Portal)
only with anomalous, excluding phases, SG 20/21, with testset & amplitudes (145/343)



For a comparison of different ready-to-use results, a user needs to fetch those 145 MTZ files (ca 7 per sweep).

That's a lot ...
Let's ask the expert ...

Useful data

User: "What MTZ file should I take?"

Expert (1990): "Have a look at the diffraction spots and if they look nice. What about spatial overlap? Mask your beamstop carefully. Pick some spots for indexing - maybe only the first few images. Check visually that all your spots are predicted. Edit the shoebox and profile parameters to catch enough background. Integrate your 90 degree of data and look at Rmerge, $\langle I/\sigma I \rangle$ and completeness. Calculate some isomorphous Patterson maps: do you have enough anomalous or isomorphous signal to find your heavy atoms on Harker sections? Have you done the self-rotation computation for NCS averaging later?"

Expert (2000): "Look at the interleaved wavelength and/or inverse-beam MAD dataset that gives you the strongest anomalous signal. Resolution is not that important since you first need to solve the structure (and collect a high-res native later anyway). See if the peak SAD dataset gives you a better set of initial HA sites. Run SHELXD over the weekend - you might only get 1 solution in a million."

Expert (2010): "Have a look at those with the highest resolution, a good Rmeas or Rpim value in the outer shell, sensible completeness and $\langle I/\sigma I \rangle$ value. For SAD data look at the anomalous signal. Be careful about radiation damage (this is why we collect high-multiplicity data). Try different pipelines."

Expert (2020): "Take the best one(s) and run MR against all possible AF models or the whole PDB - and take the best result after that."

Maybe we are still at that stage ... are we?

Expert (2026): "Take the best one."



Can we guide user to "best" MTZ file?

Auto processing ranking

We use the following criteria by order of priority:

1. Matches all set filter cutoffs
Add...
2. Highest symmetry space group
☒ Enabled
3. Selected criteria
Overall ▼ Res. high ▼

3. Selected criteria

Overall ▼ Res. high ▼

Inner	Compl.
Outer	Res. low
Overall	Res. high
	Rmeas
	<I/s(I)>
	cc1/2
	ccAno

Cutoff Configuration

	Inner	Outer	Overall
Compl. <			
Res. low <			
Res. high >			
Rmeas >			
<I/s(I)> <			
cc1/2 <			
ccAno <			

Apply Cancel

3 x 7 possibilities

Q: What settings would **you** (as an MX expert) choose to get the "best" dataset ranked highest?

=> proc-develop@globalphasing.com

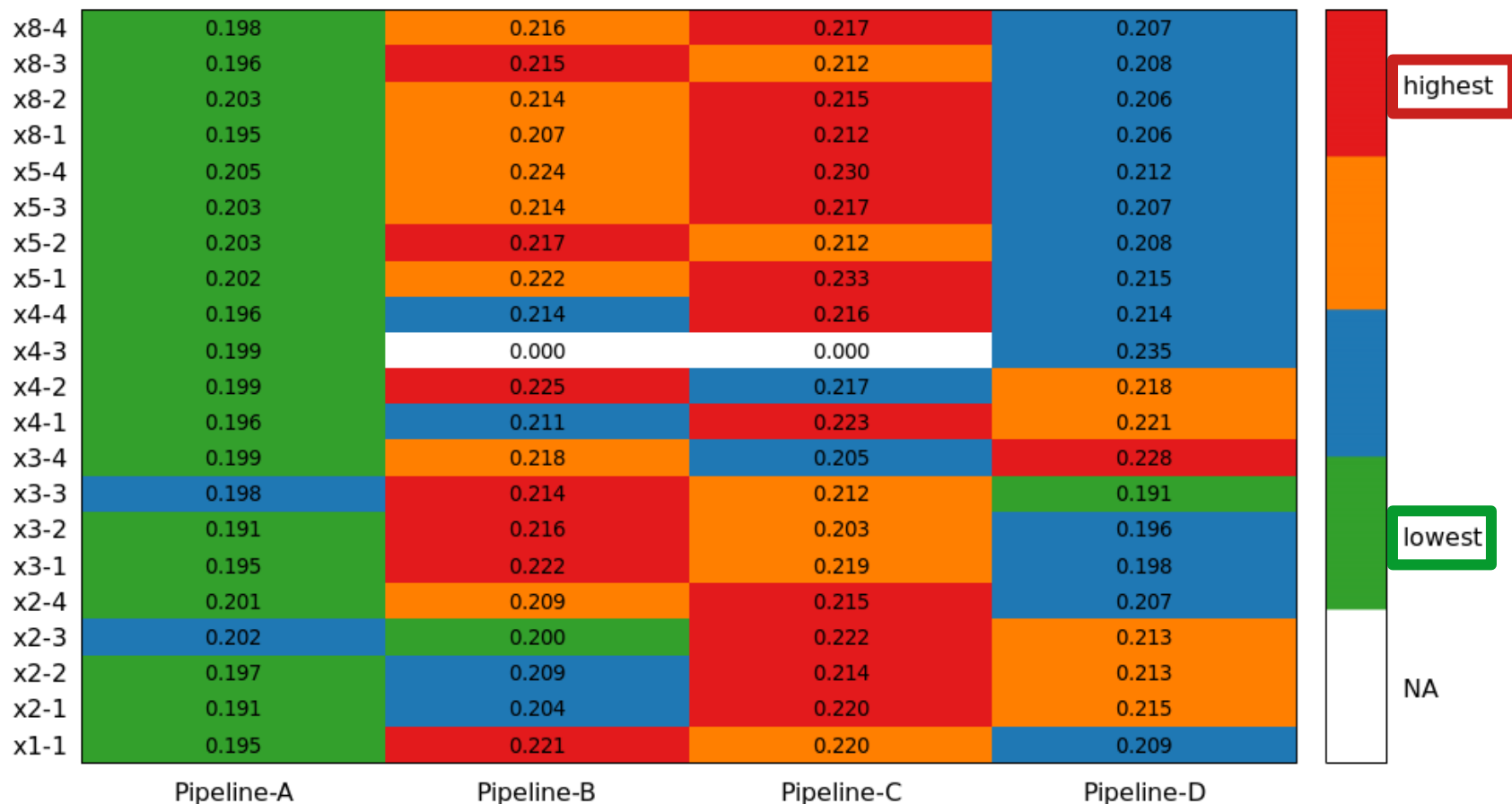
What is the "best" dataset anyway?

For the 2026/05
MASSIF-1 example

- (1) Find deposited PDB entry as a starting model.
 - **Correct SG (20)**
 - Highest resolution (to have **consistent test-set flags**)
- (2) Take all **145 merged MTZ files** (anomalous, amplitudes and **SG 20/21**).
- (3) Ensure SG 20 and consistent test-set flags.
- (4) Run "**standard**" refinements (BUSTER "refine", REFMAC "dimple").
- (5) Take **common reflections** within a sweep (Fcalc and Fobs, **scaled**).
- (6) Compute **R/Rfree** for all refinements of all datasets **for a given sweep**.
- (7) For each pipeline, take the **lowest Rfree value achieved**
- (8) This allows to **sort pipeline results** for each sweep by **Rfree** value

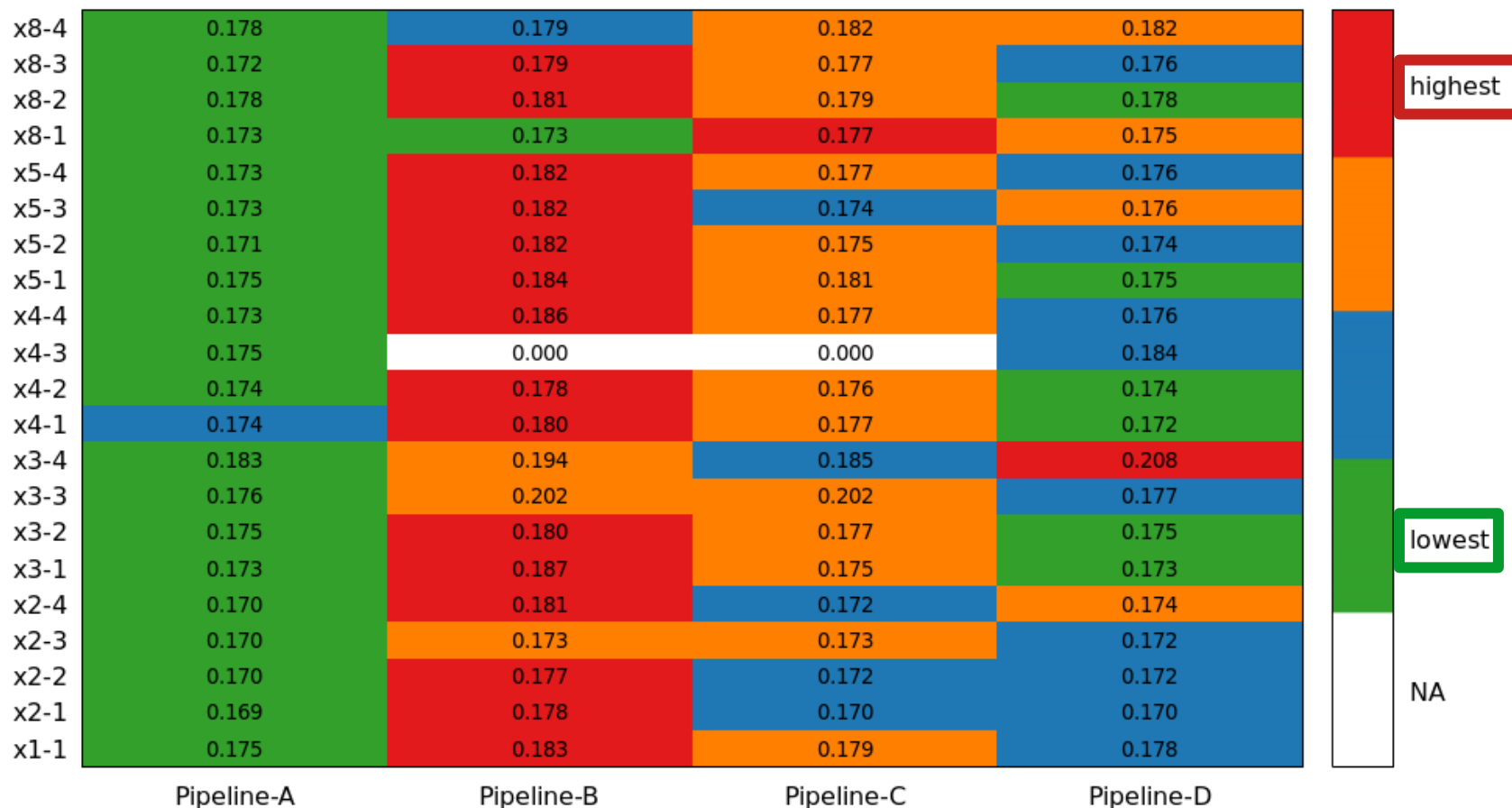
What is the "best" dataset anyway?

Rfree (common reflections) - best of each pipeline per sweep
BUSTER ranking



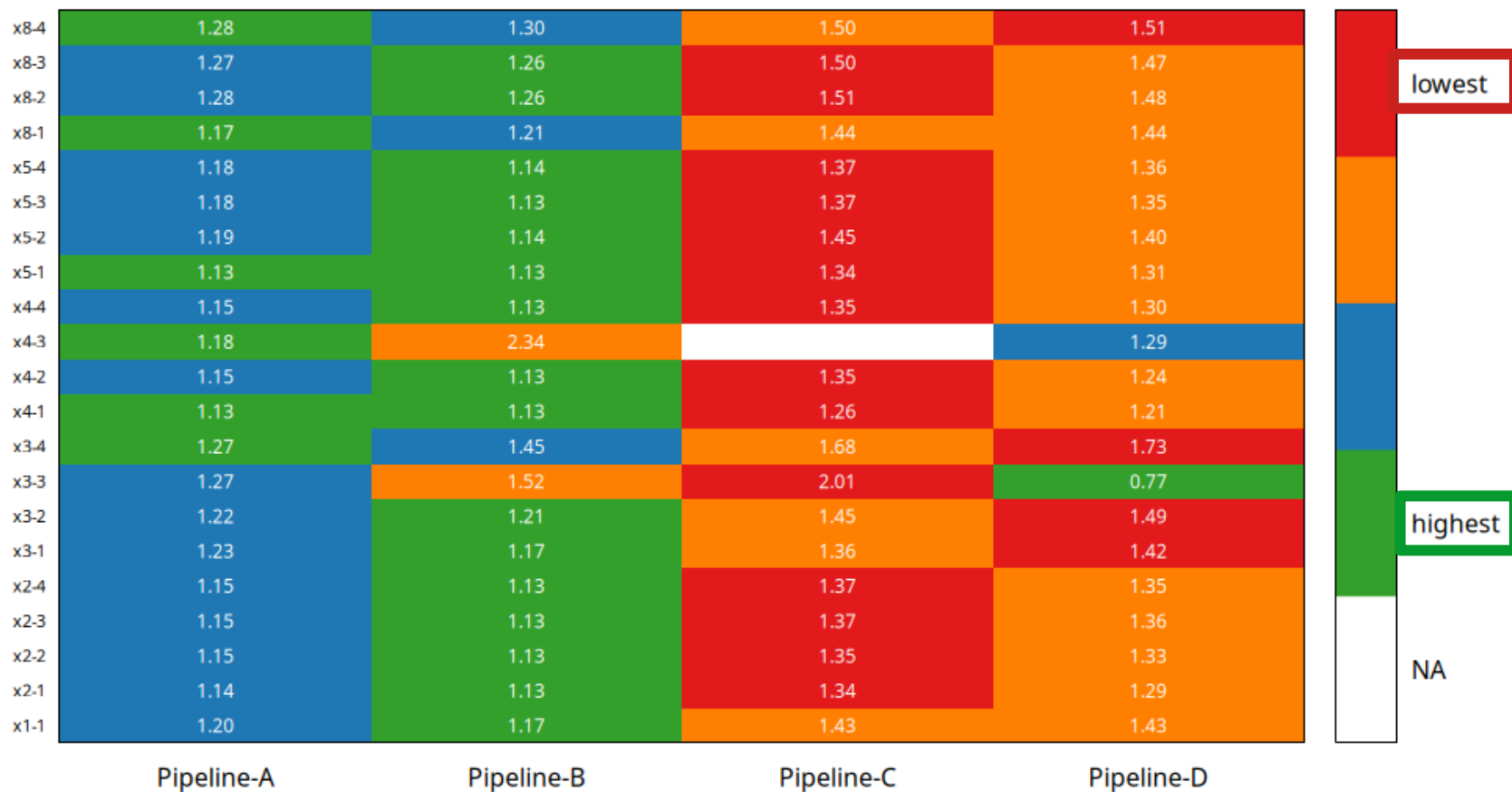
What is the "best" dataset anyway?

Rfree (common reflections) - best of each pipeline per sweep
REFMAC ranking



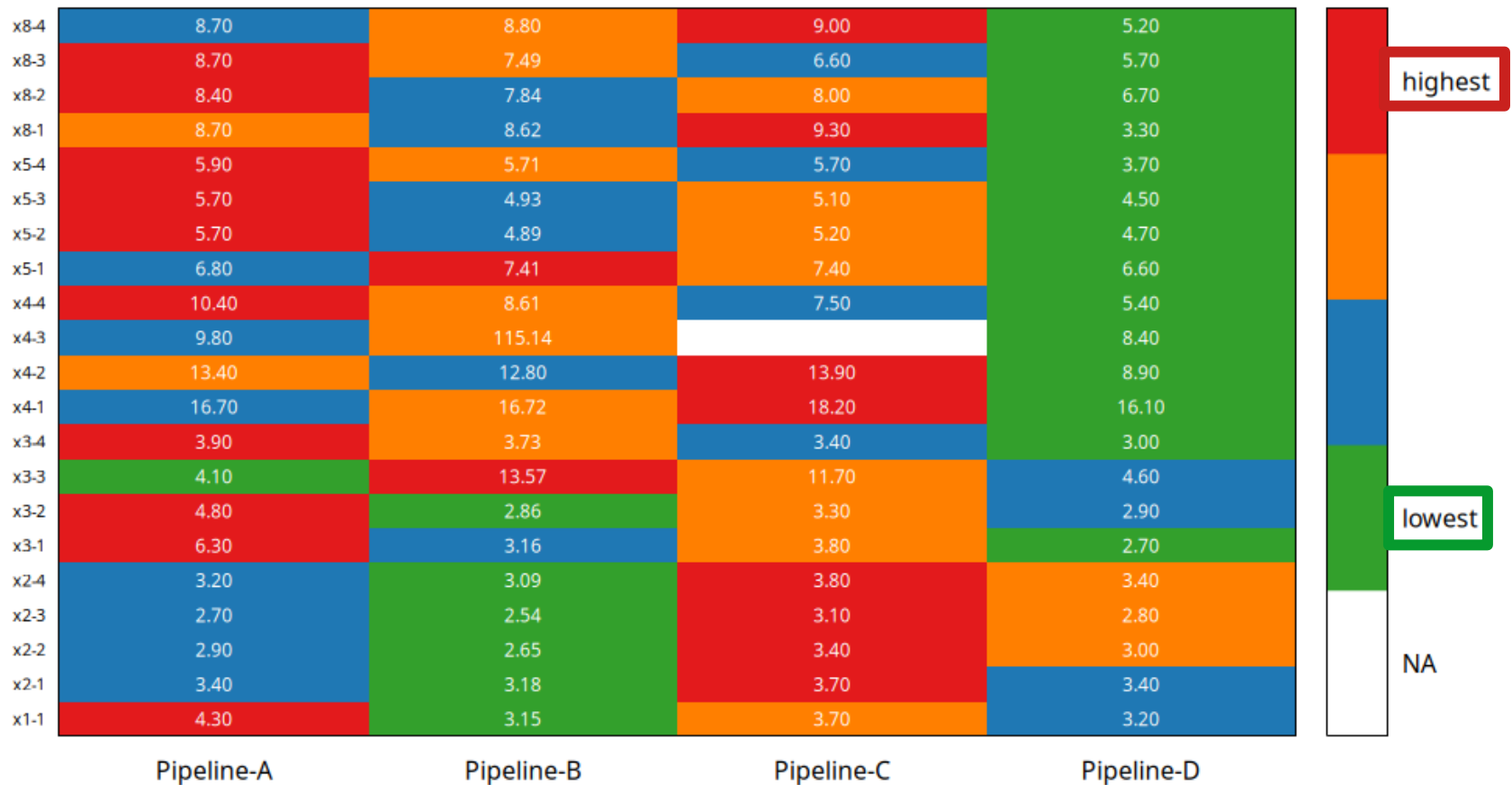
Typical (?) ranking metrics

Overall ResHigh
(best of each pipeline)



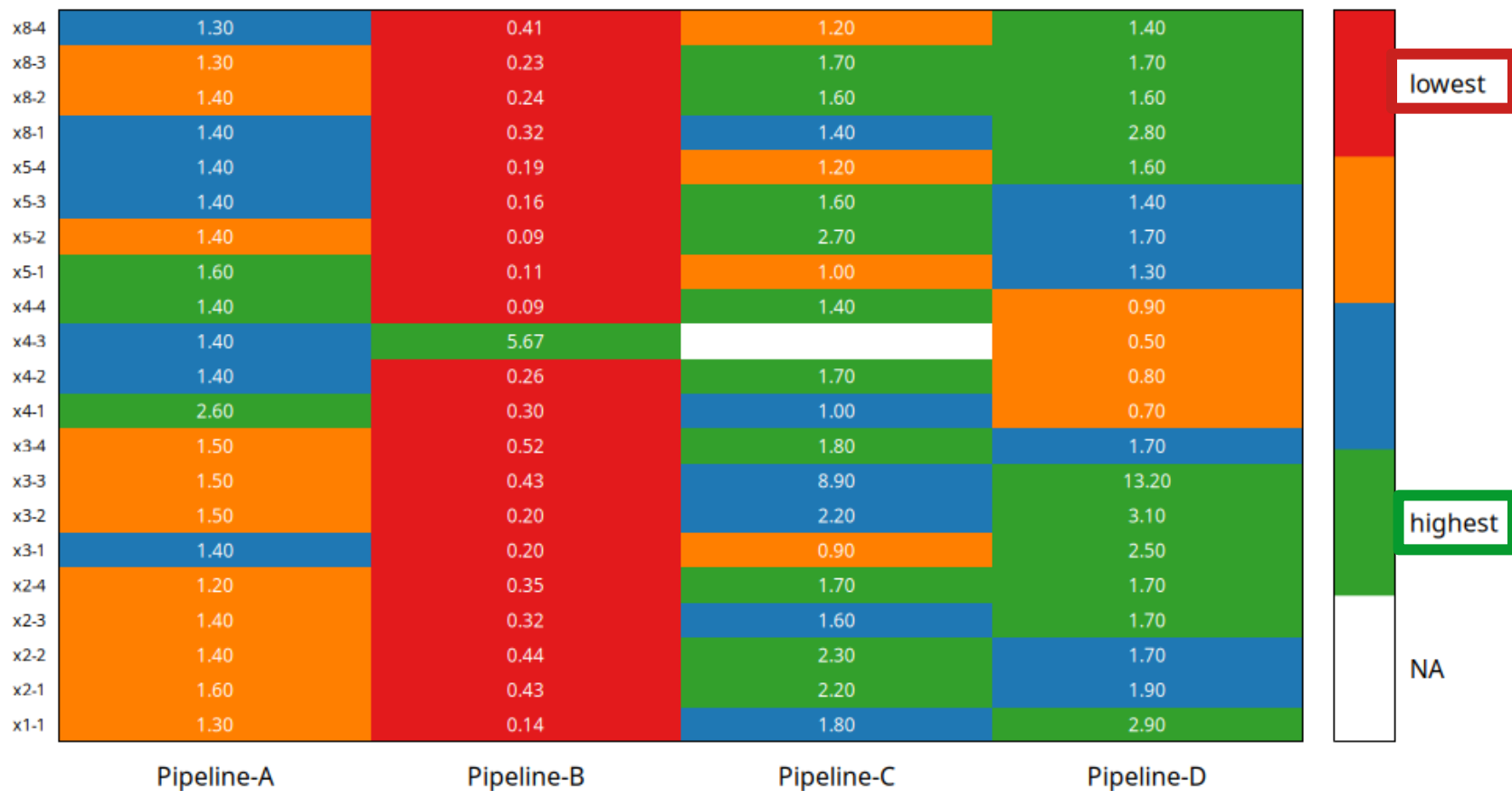
Typical (?) ranking metrics

Inner Rmeas (best of each pipeline)



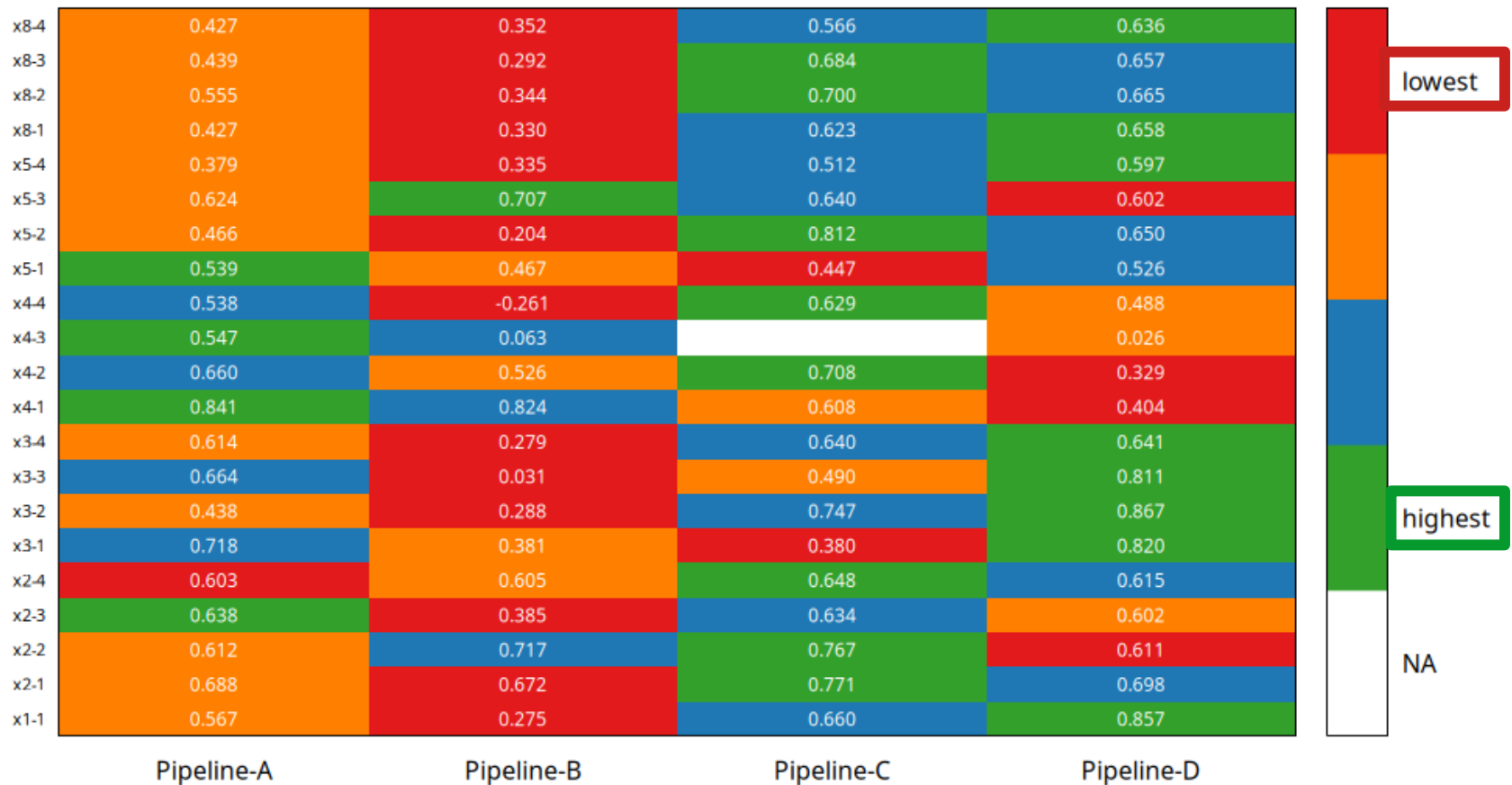
Typical (?) ranking metrics

Outer $\langle I/\sigma \rangle$
(best of each pipeline)



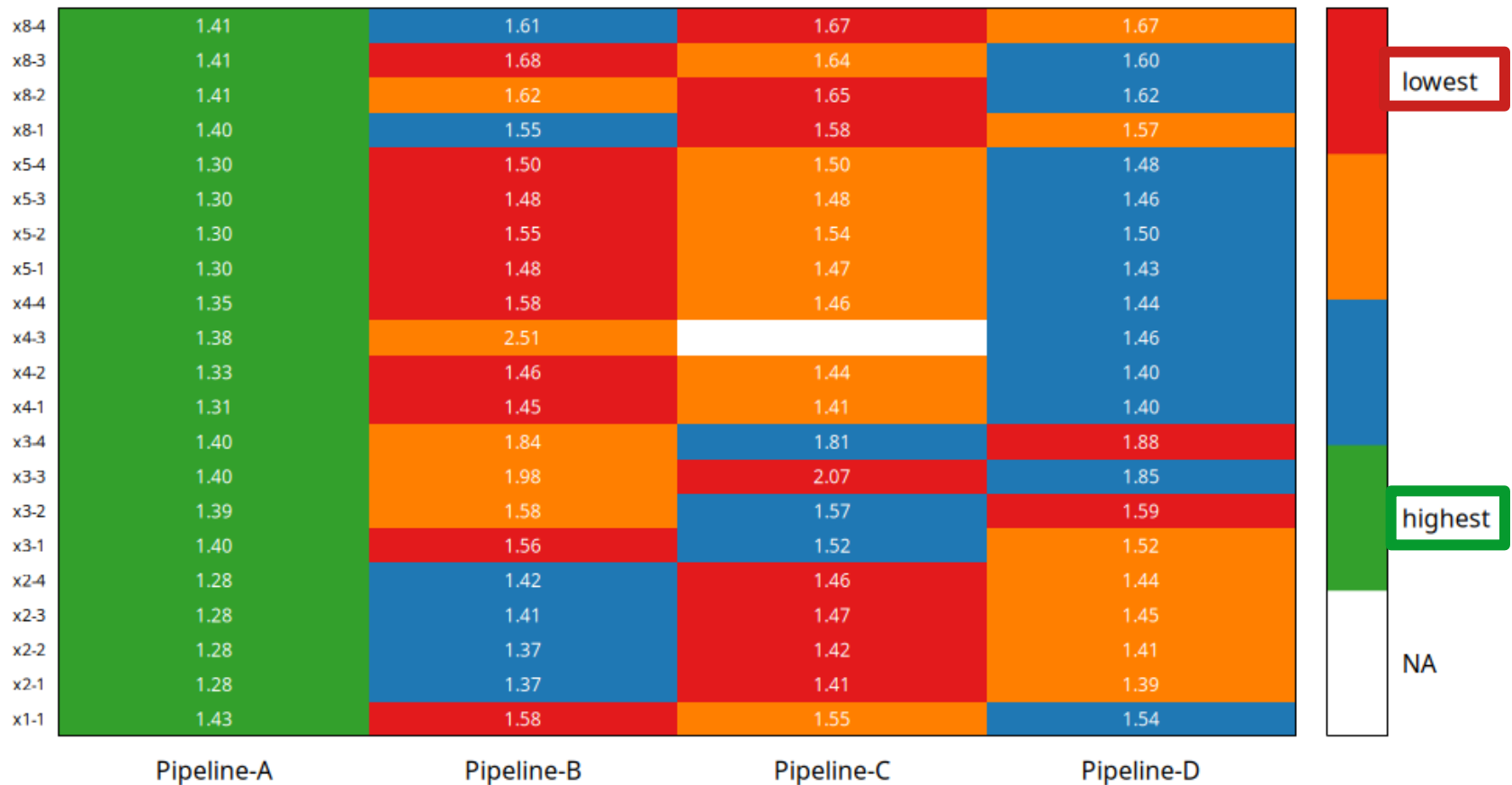
Typical (?) ranking metrics

Outer CC1/2 (best of each pipeline)



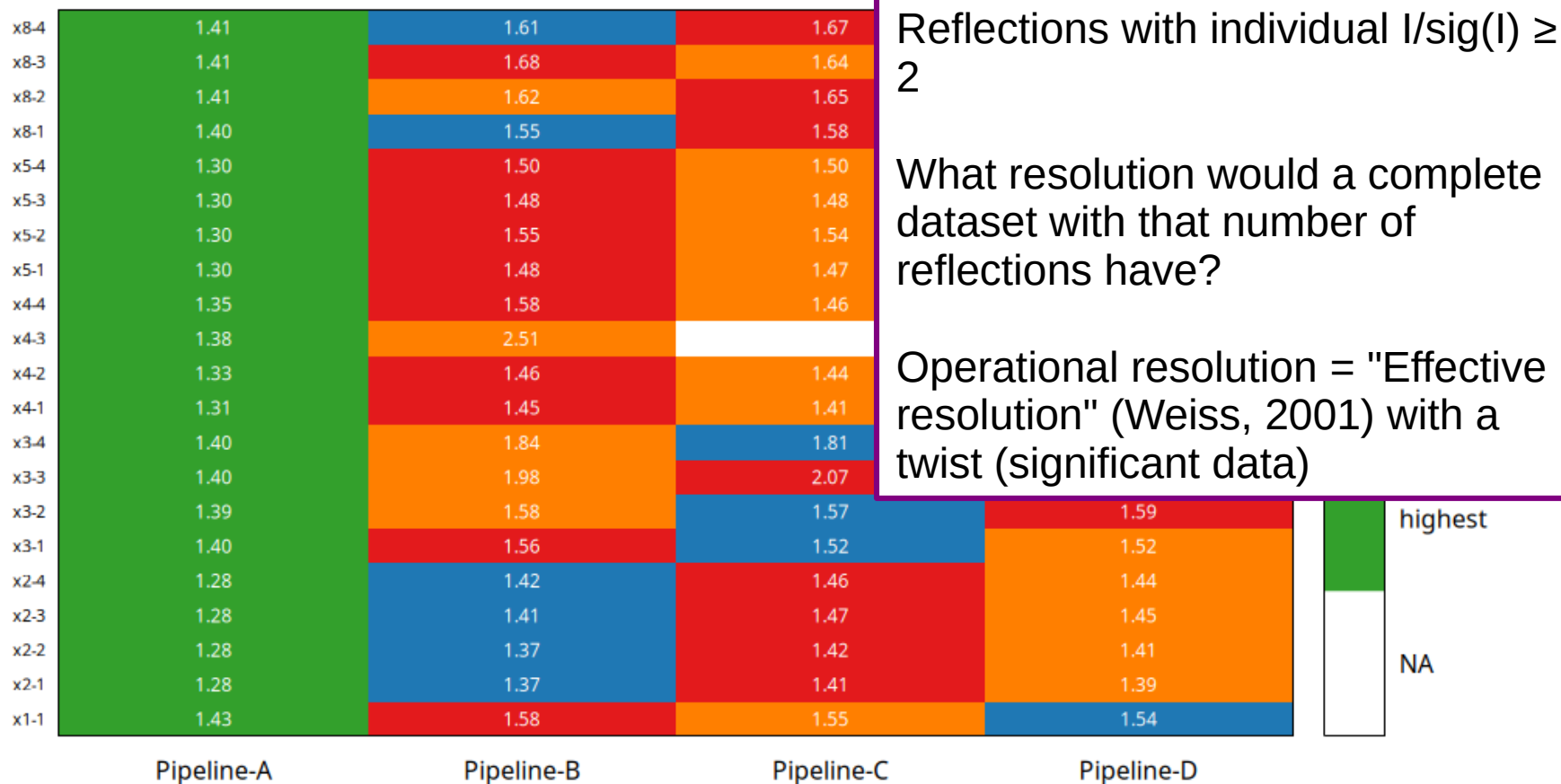
Typical (?) ranking metrics

Operational Resolution (best of each pipeline)



Typical (?) ranking metrics

Operational Resolution (best of each pipeline)



Weiss, M.S., 2001. Global indicators of X-ray data quality. Applied Crystallography, 34(2), pp.130-135.

<https://www.globalphasing.com/DQMW/> (Apr 2019, GPhL@ESRF)

Data Quality Metrics & Ranking

Q1: Are the current cutoff filtering and ranking systems fit for purpose?

Q2: Do experts use that system and are happy with the results - always using the data from the top-ranking processing job?

Q3: Are non-expert users happy with it because they know they get the best results?

Data Quality Metrics & Ranking

Q1: Are the current cutoff filtering and ranking systems fit for purpose?

Overall/Outer **metrics** can be "improved" by aggressive resolution cut.

Overall/Outer **resolution** can be "improved" by incorrect resolution cut.

Inner shell metrics are dominated by **binning** decisions

Q2: Do experts use that system and are happy with the results - always using the data from the top-ranking processing job?

Not us.

None of our power/industrial users as far as we know.

Q3: Are non-expert users happy with it because they know they get the best results?

Experience from 4-6 workshops a year around the world:

- encountering interested users - not "typical" users (biased audience)
- all users are confused every time finding their data and selecting MTZ files
- more than 3-5 choices are overkill
- users like a "best" system because they can "just move on"

Which ranking method is "best"

Assuming that the **Rfree from BUSTER or REFMAC** constitutes some kind of "ground truth" or at least a sensible reference for ranking across datasets (sweeps) and pipelines.

Use the 7 metrics across 3 shells (ranking metrics available to data portal users) plus operational resolution = **22 possibilities**

Four **agreement measures**:

- Kendall τ -b
- Spearman's rank correlation
- top1 (fraction of sweeps where the candidate's best pipeline = the reference's best)
- exact (fraction of sweeps with the full ordering identical)

Which ranking metric agrees closest with the "ground truth" (reference Rfree from BUSTER or REFMAC/dimple)?

Which ranking method is "best"

BUSTER

Candidate	tau_b	spearman	top1	exact
opres	0.740	0.815	0.905	0.381
<i>Inner-ResLow</i>	<i>0.488</i>	<i>0.539</i>	<i>0.476</i>	<i>0.238</i>
<i>Overall-ResLow</i>	<i>0.456</i>	<i>0.539</i>	<i>0.429</i>	<i>0.190</i>
Outer-Rmeas	0.358	0.443	0.333	0.143
Outer-ResHigh	0.345	0.371	0.286	0.095

Dimple/ REFMAC

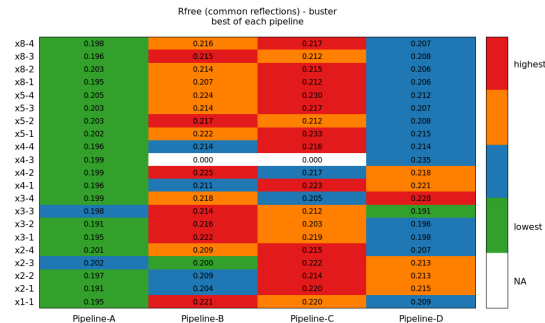
Candidate	tau_b	spearman	top1	exact
opres	0.719	0.756	0.952	0.333
Inner-CChalf	0.507	0.587	0.857	0.143
<i>Overall-ResLow</i>	<i>0.477</i>	<i>0.565</i>	<i>0.524</i>	<i>0.095</i>
Overall-CChalf	0.476	0.494	0.810	0.095
Overall-IsigI	0.330	0.386	0.524	0.143

Operational resolution (opres) predicts the refinement-based pipeline ranking far **better than any individual portal metric** - by a clear margin - *if* we are interested in a single "best". But getting the full ordering right ("exact") is still elusive.

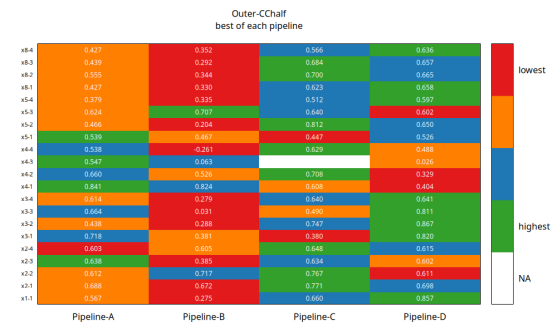
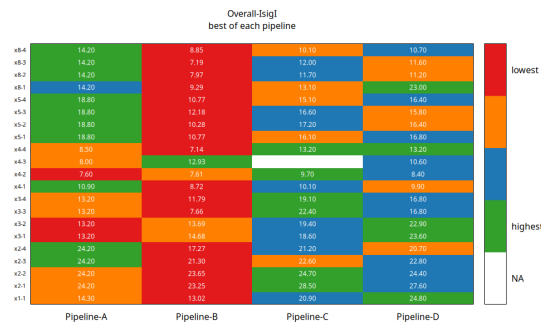
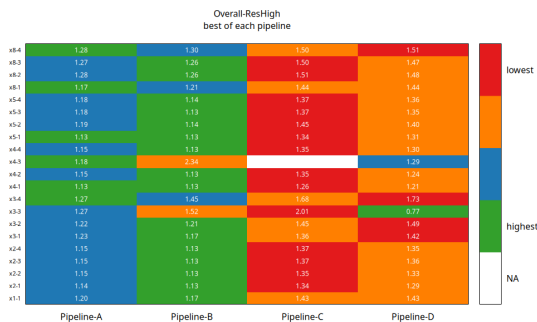
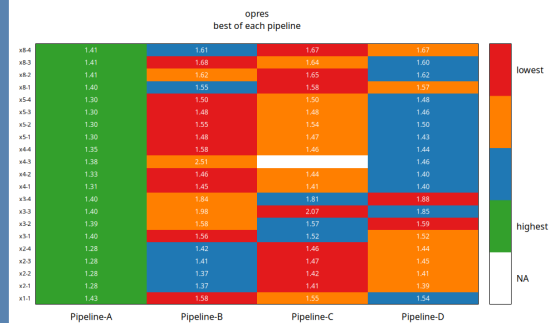
Which ranking method is "best"

Is there a single ranking metric that could give us a reasonable idea which data will perform well in refinement?

Rfree - **BUSTER**



Operational Resolution



Overall ResHigh

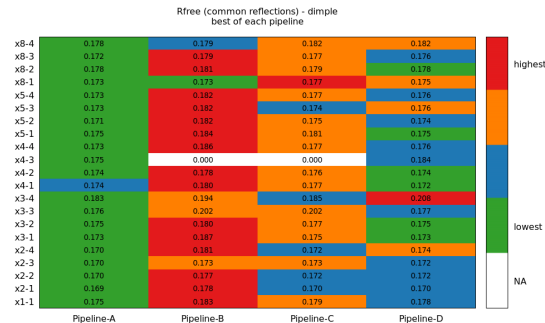
Overall <I/sigl>

Outer CC(1/2)

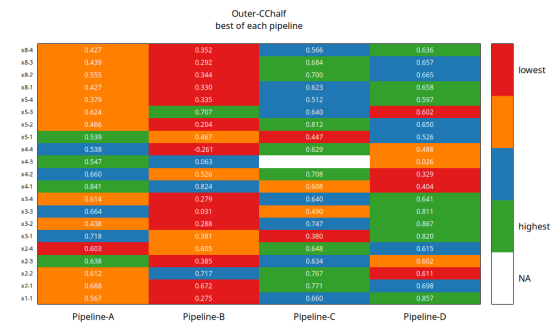
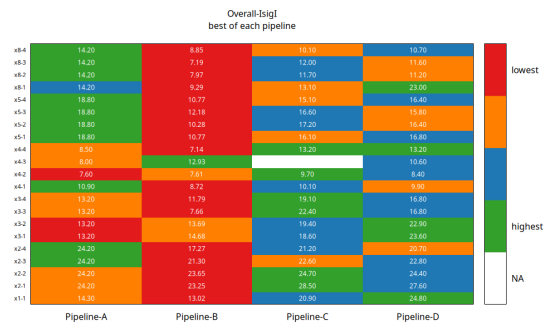
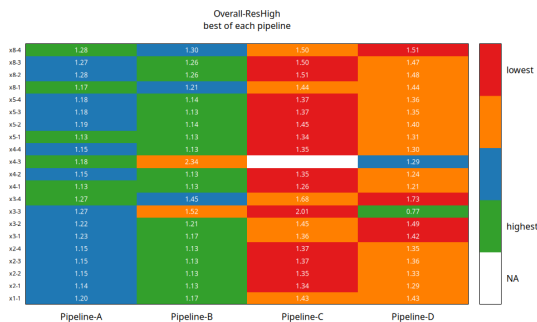
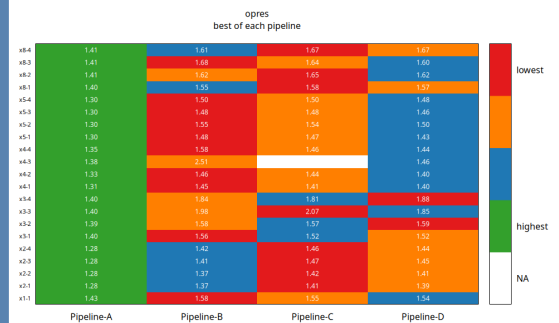
Which ranking method is "best"

Is there a single ranking metric that could give us a reasonable idea which data will perform well in refinement?

Rfree - REFMAC



Operational Resolution



Overall ResHigh

Overall $\langle 1/\sigma \rangle$

Outer CC(1/2)

Recommendations

If we want to run a very large number of pipelines for every dataset coming off a beamline, we **need some system to help users finding their way**.

This requires **real expert knowledge** to assess, highlight and **explain** the purpose as well as strengths/weaknesses of **every processing result**.

We *could* attempt to guide users to a single dataset for every collection - the so-called "best".

~~Recommendations~~

Sticky tape

If we want to run a very large number of pipelines for every dataset coming off a beamline, we **need some system to help users finding their way**.

- Individual project, crystal, dataset **support by beamline staff**.

- Extensive and always **up-to-date tutorials and documentation**.

This requires **real expert knowledge** to assess, highlight and **explain** the purpose as well as strengths/weaknesses of **every processing result**.

- Intensive **interaction with external** methods and pipeline **developers**.

- "The community" is not as strong in MX expertise as <N> decades ago.

We could attempt to guide users to a single dataset for every collection - the so-called "best".

- Investigate sensible ranking metric combinations.

- Some weighted average of ResHigh, CC1/2, <I/sigI>, Completeness ... any idea?

Recommended Recommendations

Don't run $N > 4$ processing jobs on every dataset ...

- ... just because we can (waste of CPU, CO2 budget)
- ... to cover every SG from P1 onwards "just in case"
- ... **more than 4 is impossible to keep track of**

Do not attempt to assign a "best" label to processing results ...

- ... we'll get it wrong most of the time
- ... it gives the impression of false security
- ... accuracy, precision, binning, anisotropy, signal, noise
- ... **be honest: provide results in random order**
- ... alphabetical or run-time sorting is also biased

If we REALLY still think to use a ranking metric ...

- ... use **operational resolution** (only metric that makes sense)
- ... combines completeness, signal and noise
- ... unaffected by binning, anisotropy

Acknowledgements

- Matt Bowler, Olof Svensson (ESRF)
- Gérard Bricogne, Claus Flensburg, Rasmus Fogh, Peter Keller (GPhL)

Sorry for straying away from "Multi-sweep and multi-crystal experiments" ... we covered both, but maybe not in the way originally envisaged :-)